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Predictive accuracy of a perioperative hemodynamic indices-based prediction model for moderate-to-severe acute kidney injury after orthotopic heart transplantation

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A R T I C L E I N F O

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A B S T R A C T

Background: Acute kidney injury is a common complication after orthotopic heart transplantation. Previous models have failed to consider the impact of multiple hemodynamic parameters on prediction performance. The objective of this study was aimed to develop machine-learning models incorporating multiple hemodynamic parameters to predict the risk of developing acute kidney injury after orthotopic heart transplantation.
Methods: We retrospectively analyzed 114 recipients of orthotopic heart transplantation, with postoperative stage 2–3 acute kidney injury as the prediction outcome. Preoperative characteristics, laboratory parameters, and intraoperative hemodynamics were evaluated. Potential predictors were first screened by univariate analysis, followed by least absolute shrinkage and selection operator regression for feature selection. Five machine-learning models were developed and evaluated via 5-fold cross-validation using multiple performance metrics.
Results: Postoperative stage 2–3 acute kidney injury incidence was 21.9% (25/114). Intraoperative hypotension and venous congestion were significantly associated with acute kidney injury. A total of 8 factors were ultimately filtered for constructing multiple machine-learning models, with the light gradient boosting machine model demonstrating the optimal performance (area under the receiver operating characteristic curve, 0.898; area under the precision-recall curve, 0.802; Brier score: 0.106).
Conclusion: The machine-learning model based on incorporating perioperative hemodynamics (pulmonary artery systolic pressure, mean arterial pressure, central venous pressure) accurately predict stage 2–3 acute kidney injury postorthotopic heart transplantation, thereby optimizing hemodynamic management and improving outcomes.

Introduction

Orthotopic heart transplantation (OHT) is the only therapy for patients with end-stage heart disease. With the increasing global prevalence of heart failure, the demand for OHT has continued to grow and outpace supply. In recent years, the annual number of OHTs has been relatively stable at 5,000 to 6,000 globally.[1] Although OHT technology is becoming increasingly mature, acute kidney injury (AKI) is a frequent complication after OHT. An incidence ranging from 25% to 80% significantly contributes to high morbidity and mortality.[2–4] Therefore, early identification of risk factors and active intervention are imperative to improve the long-term survival and quality of life of patients who undergo OHT.

Patients awaiting OHT typically have compromised hemodynamic reserve as a result of end-stage heart failure.[5,6] At the same time, the kidney is one of the most sensitive organs to hemodynamic changes, and its function is highly dependent on stable

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perfusion pressure and adequate blood supply.[7,8] It has been shown that the primary causes of AKI after heart transplantation include renal hemodynamic disturbances, nephrotoxic-free hemoglobin, inflammation and oxidative stress, and microembolism.[9,10] Among these, hemodynamic instability plays a significant role in kidney injury associated with cardiac surgery.[11–13] Interventions like goal-directed perfusion and remote ischemic preconditioning improve renal hemodynamics, thereby reducing AKI risk and improving patient outcomes.[14] Therefore, optimal hemodynamic management is essential to maintaining an adequate glomerular pressure gradient to protect renal function. Although numerous studies have explored the impact of various perioperative factors on AKI, no research has yet investigated whether pulmonary artery systolic pressure (PASP), central venous pressure (CVP), and intraoperative mean arterial pressure (MAP) can collectively predict the occurrence and progression of moderate-to-severe AKI after OHT.

The primary objective of this study was to develop and validate machine learning (ML)-based predictive models for early AKI risk stratification, assessing their precision in perioperative clinical decision-making. The secondary objective was to investigate the impact of hemodynamic alterations during sequential phases of OHT surgery, specifically pre-cardiopulmonary bypass (CPB), intra-CPB, and post-CPB periods, on the development of stage 2–3 AKI. These findings are expected to establish a scientific basis for optimizing perioperative management and improving patient prognosis.

Methods

Patient population

The patient population originated from a cross-sectional study at the Nanjing First Hospital between October 2011 and September 2023. The inclusion criteria were adult patients who had undergone OHT surgery. Patients were excluded if they met the following criteria: (1) lack of perioperative serum creatinine (Scr) data; (2) with preoperative AKI, end-stage renal disease, or dialysis; (3) with re-transplantation or other combined organ transplants; (4) death within 48 hours after surgery.

The study protocol complied with the Declaration of Helsinki and received approval from the Nanjing First Hospital Ethics Committee (Nanjing, Jiangsu, China; approval number: KY202 21122-01-KS-01). As the study was retrospective, the ethics committee waived the requirement for informed consent. The study design is outlined in Figure 1.

Procedures

All recipients underwent OHT with double-chamber donor implantation under general anesthesia. Anesthesia was maintained with sevoflurane before and after CPB and propofol during CPB. The preservation solution utilized for all transplants was the University of Wisconsin solution at a volume of 1,000 cc. Pulmonary artery pressure was continuously monitored using a Swan-Ganz floating catheter inserted via the right internal jugular vein puncture. In contrast, invasive arterial pressure was assessed through a left radial artery puncture. Postoperatively, patients received ventilator-assisted breathing and vasoactive agents such as dopamine, epinephrine, and norepinephrine to ensure hemodynamic stability. Patients exhibiting postoperative pulmonary hypertension (pulmonary systolic pressure >40 mm Hg [1 mm Hg = 0.133 kPa]) were administered intravenous milrinone infusion. After surgery, piperacillin and tazobactam sodium were used to prevent bacterial infection, along with a triple immunosuppressive regimen consisting of motephenate, tacrolimus, and glucocorticoids.

Data collection

This study collected patient demographics, comorbidities and medical history; preoperative laboratory and echocardiography data; and intraoperative and postoperative data from the electronic medical records. Patient demographic variables included age, sex, body mass index, smoking, American Society of Anesthesiologists, and New York Heart Association functional class. The comorbidities and medical history included diabetes, hypertension, chronic liver disease, chronic kidney disease (defined by Kidney Disease: Improving Global Outcomes [KDIGO] 2012 criteria as either kidney damage or eGFR <60 mL/min/1.73 m^2 persisting ≥3 months), coronary atherosclerotic heart disease, history of stroke or transient ischemic attack, and history of previous cardiac surgery. Preoperative laboratory data included Scr, aspartate transaminase, hemoglobin, blood glucose, N-terminal pro-brain natriuretic peptide, neutrophil, lymphocyte and neutrophil to lymphocyte ratio. Preoperative echocardiography data included left ventricular ejection fraction, PASP, pulmonary artery diastolic pressure, left atrial anteroposterior dimension (LAD), and right atrial long axis dimension. Intraoperative variables included the duration of the surgery, CPB and aortic crossclamping (ACC), blood loss, urine output, colloid and crystalloid infusion, transfusion of red blood cell, blood platelet, cryoprecipitate, and autologous blood, use of diuretics and vasoactive-inotropic score (VIS). Postoperative variables included hemoglobin, neutrophil, lymphocyte, neutrophil to lymphocyte ratio, and VIS measured within 24 hours after surgery. The VIS was calculated using the following variables: (1) dopamine dose (μg/kg/min); (2) dobutamine dose (μg/kg/min); (3) epinephrine dose (μg/kg/min); (4) milrinone dose (μg/kg/min); (5) vasopressin dose (U/kg/min); (6) norepinephrine dose (μg/kg/min); and (7) levosimendan dose (μg /kg/min). The formula is as follows[15]:

$$\begin{aligned} VIS = {} & \text{dopamine} + \text{dobutamine} + 100 \times \text{epinephrine} + 10 \\ & \times \text{milrinone} + 10\,000 \times \text{vasopressin} + 100 \\ & \times \text{norepinephrine} + 50 \times \text{levosimendan} \end{aligned}$$

In addition, intraoperative blood pressure time-series data recorded every 5 minutes were exported from the Anesthesia Information Management System and subjected to secondary feature transformation to reflect the patient's intraoperative hemodynamics. Thresholds for MAP (<55, <60, <65 mm Hg) and CVP (>12, >16, >20 mm Hg) were set due to published variability in intraoperative hypotension and venous congestion.[16,17] Furthermore, in consideration of the impact of CPB on hemodynamics, the intraoperative period was divided into multiple periods, including the entire surgical procedure, from the start of the surgery to the initiation of CPB, throughout the CPB, and from the end of CPB to the end of surgery. The patient's MAP and CVP duration at a given threshold and the total area under the curve (AUT) calculated on the basis of the trapezoidal rule were calculated separately for each of the aforementioned multiple time periods.[18,19] The calculation methodology is illustrated in the Supplementary Figure.

Outcome

The predicted outcome was the occurrence of postoperative stage 2–3 AKI. According to the KDIGO Clinical Practice Guidelines, stage 2–3 AKI is a 2-fold or greater increase in Scr within the first 7 days after surgery,[20] the baseline Scr was determined by the

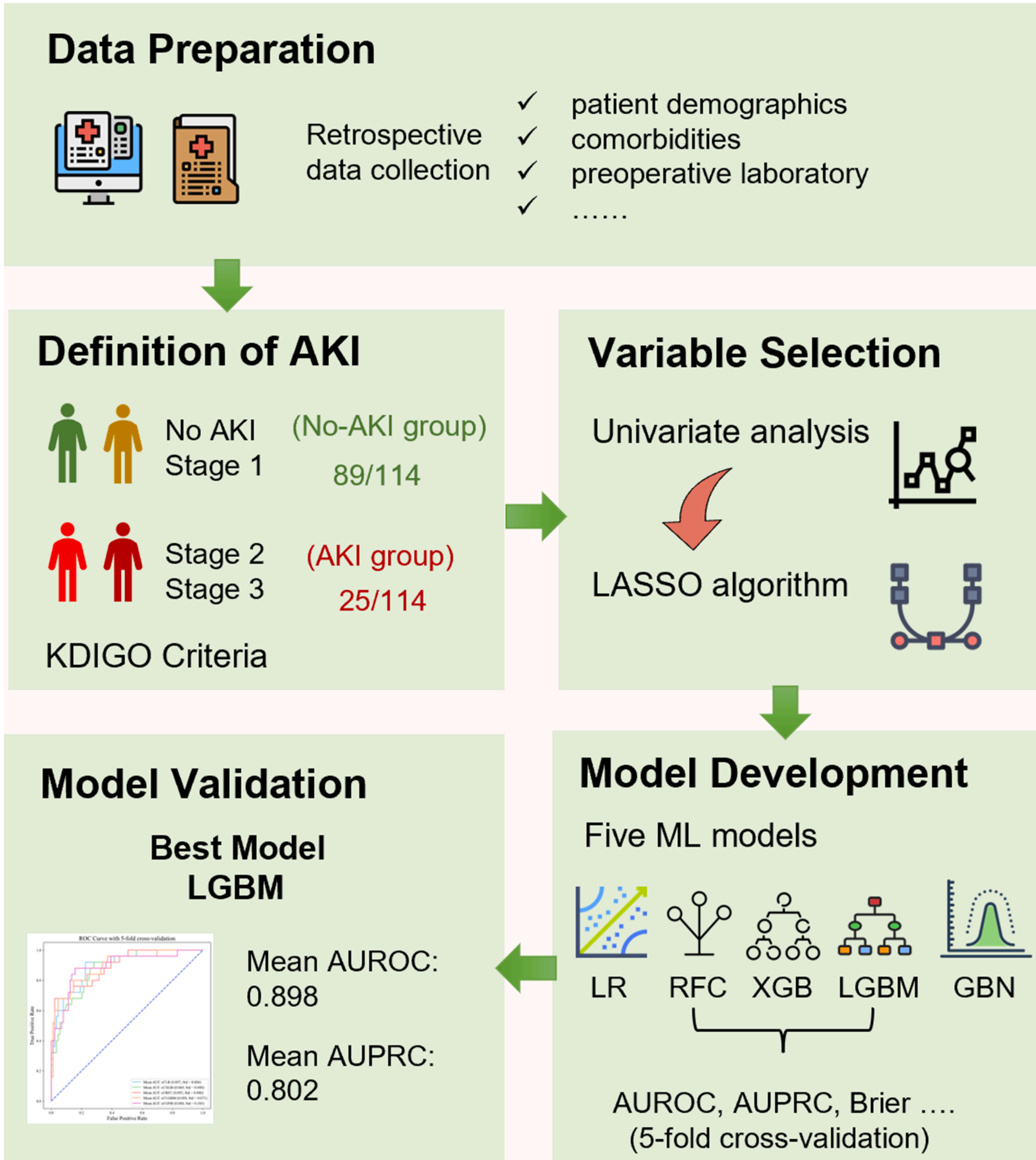

Figure 1. Flowchart of the study design. *AKI*, acute kidney injury; *AUPRC*, the area under the precision-recall (PR) curve; *AUROC*, the area under the receiver operating characteristic curve; *GNB*, Gaussian naive Bayes; *KDIGO*, Kidney Disease: Improving Global Outcomes; *LASSO*, least absolute shrinkage and selection operator; *LGBM*, light gradient boosting machine; *LR*, logistic regression; *RFC*, random forest classifier; *XGB*, extreme gradient boosting.

most recent concentration measurement obtained within 48 hours before surgery.

Data preprocessing and variable selection

Initially, variables with a missing rate exceeding 20% were excluded from the dataset. Variables below the threshold were interpolated using the K-nearest neighbor method and retained for subsequent analysis.[21] Subsequently, univariate analysis was employed to identify variables associated with postoperative stage 2–3 AKI ($P < .05$). The least absolute shrinkage and selection operator (LASSO) algorithm was then used to screen the independent predictors for stage 2–3 AKI. The LASSO algorithm uses the hyperparameter lambda (λ) to minimize the regression coefficients toward zero during the model estimation process.[22] This approach excludes many weakly correlated features by assigning their zero-value coefficients. Only those variables with a non-zero coefficient were selected for further analysis. Furthermore, we also estimated the multicollinearity of the variables chosen using the variance inflation factor (VIF), and VIF <5 was considered insignificant.[23] Finally, each continuous variable was subjected to z-score normalization, while each categorical variable underwent one-hot encoding.

Model development and evaluation

In this study, we developed and validated 5 different ML models to predict stage 2–3 AKI, including logistic regression, extreme gradient boosting, random forest classifier, light gradient boosting machine (LGBM), and Gaussian naive Bayes models. The grid search algorithm and 5-fold cross-validation (CV) were used to optimize each model's hyperparameters. The "sklearn 1.5.2,"

Table I
The demographic and clinical characteristics in the whole cohort and potential risk factors related to 2 or 3 AKI by univariate analysis

	Total (*n* = 114)	Non-AKI stage 2 or 3 (*n* = 89)	AKI stage 2 or 3 (*n* = 25)	*P* value
Demographic				
Age, yr	52.00 [45.00, 60.00]	51.00 [45.00, 59.00]	53.00 [49.00, 60.00]	.476
Sex = male	93 (81.6)	74 (83.1)	19 (76.0)	.601
BMI, kg/m^2	22.96 (3.41)	23.05 (3.57)	22.64 (2.81)	.593
Smoking				.458
Never	71 (62.3)	57 (64.0)	14 (56.0)	
ormer	21 (18.4)	17 (19.1)	4 (16.0)	
Current	22 (19.3)	15 (16.9)	7 (28.0)	
ASA				.772
III	25 (21.9)	20 (22.5)	5 (20.0)	
IV	76 (66.7)	58 (65.2)	18 (72.0)	
V	13 (11.4)	11 (12.4)	2 (8.0)	
NYHA functional class				.617
III	43 (37.7)	32 (36.0)	11 (44.0)	
IV	71 (62.3)	57 (64.0)	14 (56.0)	
Comorbidities and medical history				
Diabetes = yes	30 (26.3)	24 (27.0)	6 (24.0)	.968
Hypertension = yes	29 (25.4)	21 (23.6)	8 (32.0)	.553
Chronic liver disease = yes	6 (5.3)	5 (5.6)	1 (4.0)	.999
Chronic kidney disease = yes	10 (8.8)	5 (5.6)	5 (20.0)	.065
Coronary atherosclerotic heart disease = yes	23 (20.2)	14 (15.7)	9 (36.0)	.051
History of stroke or TIA = yes	8 (7.0)	6 (6.7)	2 (8.0)	.999
History of previous cardiac surgery = yes	47 (41.2)	32 (36.0)	15 (60.0)	.054
Preoperative laboratory data				
Scr, μmol/L	89.60 [73.30, 114.10]	88.90 [72.90, 106.00]	98.00 [75.30, 145.90]	.114
eGFR, mL/min/1.73 m^2	63.77 [48.72, 77.74]	64.19 [51.40, 77.75]	57.16 [36.26, 74.63]	.086
AST, U/L	25.50 [22.00, 34.75]	25.00 [21.00, 31.00]	27.00 [23.00, 41.00]	.124
Hemoglobin, g/L	131.00 [118.00, 146.75]	134.00 [122.00, 151.00]	109.00 [82.00, 122.00]	**<.001**[*]
Blood glucose, mmol/L	5.96 [5.24, 7.20]	5.85 [5.22, 7.03]	6.14 [5.51, 7.40]	.278
NT-proBNP, pg/mL	10,112.67 [3,628.63, 11,636.96]	9,394.15 [2,951.78, 11,552.34]	10,812.82 [9,435.16, 12,131.99]	**.041**[*]
Neutrophil, %	72.12 (11.00)	71.18 (10.38)	75.48 (12.63)	.084
Lymphocyte, %	19.44 (9.32)	20.37 (8.77)	16.14 (10.63)	**.044**[*]
NLR, %	3.48 [2.51, 5.79]	3.39 [2.41, 5.03]	5.23 [2.67, 11.89]	.077
Preoperative echocardiography data				
LVEF, %	27.00 [22.25, 31.00]	26.00 [22.00, 30.00]	29.00 [25.00, 32.00]	.135
PASP, mm Hg	44.00 [38.00, 50.00]	42.00 [36.00, 48.00]	51.00 [43.00, 56.00]	**.001**[*]
PADP, mm Hg	25.00 [21.00, 28.00]	25.00 [20.00, 28.00]	26.00 [24.00, 30.00]	**.028**[*]
LAD	52.00 [48.00, 59.00]	50.00 [48.00, 55.00]	60.00 [56.00, 64.00]	**<.001**[*]
Right atrial long axis dimension, mm	52.00 [46.00, 61.00]	50.00 [46.00, 57.00]	62.00 [58.00, 68.00]	**<.001**[*]
Intraoperative data				
Surgery duration, min	310.00 [290.00, 363.75]	305.00 [290.00, 355.00]	345.00 [310.00, 400.00]	**.005**[*]
CPB duration, min	170.00 [153.25, 190.00]	164.00 [151.00, 184.00]	188.00 [166.00, 221.00]	**.002**[*]
ACC duration, min	91.00 [79.00, 103.75]	88.00 [78.00, 102.00]	100.00 [90.00, 128.00]	**.002**[*]
Blood loss, mL	1,225.00 [900.00, 1,500.00]	1,200.00 [900.00, 1,500.00]	1,400.00 [1,050.00, 1,850.00]	**.034**[*]
Urine output, mL/kg/h	1.92 [1.34, 2.77]	2.19 [1.57, 3.05]	1.18 [0.26, 1.94]	**<.001**[*]
Colloid infusion, mL/kg/h	2.50 [1.83, 3.25]	2.60 [1.85, 3.42]	2.19 [1.72, 3.03]	.138
Crystalloid infusion, mL/kg/h	2.26 [1.65, 3.20]	2.32 [1.71, 3.43]	2.05 [1.51, 2.77]	.181
Red blood cell transfusion, units	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	**.011**[*]
Blood platelet transfusion, units	2.00 [1.00, 2.00]	2.00 [1.00, 2.00]	2.00 [1.00, 2.00]	.451
Cryoprecipitation transfusion, units	5.85 [0.00, 11.25]	7.00 [0.00, 11.25]	4.50 [0.00, 9.75]	.185
Autologous blood transfusion, ml	750.00 [500.00, 1,000.00]	750.00 [500.00, 1,000.00]	750.00 [500.00, 1,000.00]	.260
Diuretics use = yes	37 (32.5)	27 (30.3)	10 (40.0)	.503
VIS	9.00 [5.00, 17.75]	8.00 [5.00, 16.00]	14.00 [8.00, 24.75]	.021[*]
Hemodynamic data				
Hypotension during surgery				
MAP <65 time, min	165.00 [110.00, 230.00]	155.00 [100.00, 200.00]	230.00 [200.00, 290.00]	<.001[*]
MAP <60 time, min	112.50 [65.00, 178.75]	95.00 [60.00, 150.00]	185.00 [150.00, 220.00]	<.001[*]
MAP <55 time, min	65.00 [35.00, 125.00]	55.00 [35.00, 110.00]	125.00 [95.00, 140.00]	.001[*]
MAP_AUT_65, min·mm Hg	1,850.00 [1,006.25, 2,778.75]	1,452.50 [930.00, 2,415.00]	2,730.00 [2,145.00, 3,417.50]	.001[*]
MAP_AUT_60, min·mm Hg	1,010.75 [527.38, 1726.88]	832.50 [512.50, 1,457.50]	1,575.00 [1,175.00, 2,142.50]	.004[*]
MAP_AUT_55, min·mm Hg	475.00 [232.50, 840.00]	390.00 [215.00, 782.50]	737.50 [480.00, 1,265.00]	.009[*]
Hypotension during CPB				
CPB_MAP <65 time, min	115.00 [75.00, 155.00]	105.00 [70.00, 140.00]	150.00 [110.00, 190.00]	.011[*]
CPB_MAP <60 time, min	80.00 [45.00, 135.00]	70.00 [45.00, 120.00]	120.00 [55.00, 150.00]	.087
CPB_MAP <55 time, min	50.00 [25.00, 105.00]	45.00 [20.00, 90.00]	90.00 [30.00, 120.00]	.286
CPB_MAP_AUT_65, min·mm Hg	1,061.25 [581.25, 2,183.75]	1,027.50 [557.50, 1,925.00]	1,830.00 [925.00, 2,275.00]	.141
CPB_MAP_AUT_60, min·mm Hg	651.25 [290.00, 1,348.75]	582.50 [252.50, 1,215.00]	1,140.00 [372.50, 1,390.00]	.329
CPB_MAP_AUT_55, min·mm Hg	300.00 [101.88, 690.62]	295.00 [95.00, 682.50]	505.00 [122.50, 715.00]	.567
Hypotension pre_CPB				
Pre_CPB_MAP <65 time, min	10.00 [0.00, 20.00]	10.00 [0.00, 20.00]	15.00 [0.00, 25.00]	.331
Pre_CPB_MAP <60 time, min	5.00 [0.00, 10.00]	5.00 [0.00, 10.00]	0.00 [0.00, 10.00]	.977

(*continued on next page*)

Table I (*continued*)

	Total ($n = 114$)	Non-AKI stage 2 or 3 ($n = 89$)	AKI stage 2 or 3 ($n = 25$)	*P* value
Pre_CPB_MAP <55 time, min	0.00 [0.00, 5.00]	0.00 [0.00, 5.00]	0.00 [0.00, 5.00]	.753
Pre_CPB_MAP_AUT_65, min·mm Hg	26.25 [0.00, 101.25]	25.00 [0.00, 72.50]	37.50 [0.00, 137.50]	.599
Pre_CPB_MAP_AUT_60, min·mm Hg	5.00 [0.00, 29.38]	5.00 [0.00, 27.50]	0.00 [0.00, 52.50]	.954
Pre_CPB_MAP_AUT_55, min·mm Hg	0.00 [0.00, 7.50]	0.00 [0.00, 7.50]	0.00 [0.00, 15.00]	.691
Hypotension post_CPB				
Post_CPB_MAP <65 time, min	30.00 [10.00, 70.00]	30.00 [10.00, 55.00]	65.00 [20.00, 100.00]	.005*
Post_CPB_MAP <60 time, min	10.00 [0.00, 50.00]	10.00 [0.00, 35.00]	40.00 [10.00, 80.00]	.009*
Post_CPB_MAP <55 time, min	0.00 [0.00, 20.00]	0.00 [0.00, 15.00]	10.00 [0.00, 80.00]	.013*
Post_CPB_MAP_AUT_65, min·mm Hg	345.00 [163.12, 711.88]	302.50 [140.00, 557.50]	675.00 [222.50, 1,610.00]	.006*
Post_CPB_MAP_AUT_60, min·mm Hg	165.00 [73.12, 297.50]	137.50 [67.50, 260.00]	297.50 [122.50, 1,142.50]	.009*
Post_CPB_MAP_AUT_55, min·mm Hg	66.25 [25.00, 132.50]	52.50 [20.00, 115.00]	122.50 [52.50, 465.00]	.011*
Venous congestion during surgery				
CVP >12 time, min	20.00 [0.00, 80.00]	20.00 [0.00, 65.00]	40.00 [0.00, 120.00]	.121
CVP >16 time, min	0.00 [0.00, 15.00]	0.00 [0.00, 10.00]	5.00 [0.00, 55.00]	.055
CVP >20 time, min	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	.441
CVP_AUT_12, min·mm Hg	76.25 [0.00, 305.00]	52.50 [0.00, 210.00]	222.50 [0.00, 560.00]	.066
CVP_AUT_16, min·mm Hg	0.00 [0.00, 89.38]	0.00 [0.00, 82.50]	27.50 [0.00, 425.00]	.046*
CVP_AUT_20, min·mm Hg	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	.428
Venous congestion during CPB				
CPB_CVP >12 time, min	0.00 [0.00, 5.00]	0.00 [0.00, 5.00]	0.00 [0.00, 5.00]	.450
CPB_CVP >16 time, min	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	.625
CPB_CVP >20 time, min	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	.441
CPB_CVP_AUT_12, min·mm Hg	0.00 [0.00, 9.38]	0.00 [0.00, 10.00]	0.00 [0.00, 7.50]	.667
CPB_CVP_AUT_16, min·mm Hg	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	.561
CPB_CVP_AUT_20, min·mm Hg	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	.428
Venous congestion pre_CPB				
Pre_CPB_CVP >12 time, min	0.00 [0.00, 13.75]	0.00 [0.00, 10.00]	5.00 [0.00, 20.00]	.029*
Pre_CPB_CVP >16 time, min	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 5.00]	.307
Pre_CPB_CVP >20 time, min	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	.118
Pre_CPB_CVP_AUT_12, min·mm Hg	0.00 [0.00, 30.00]	0.00 [0.00, 17.50]	17.50 [0.00, 72.50]	.027*
Pre_CPB_CVP_AUT_16, min·mm Hg	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 7.50]	.283
Pre_CPB_CVP_AUT_20, min·mm Hg	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	.107
Venous congestion post_CPB				
Post_CPB_CVP >12 time, min	15.00 [0.00, 60.00]	15.00 [0.00, 50.00]	25.00 [0.00, 95.00]	.310
Post_CPB_CVP >16 time, min	0.00 [0.00, 5.00]	0.00 [0.00, 5.00]	5.00 [0.00, 45.00]	.030*
Post_CPB_CVP >20 time, min	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 15.00]	.002*
Post_CPB_CVP_AUT_12, min·mm Hg	38.75 [0.00, 183.12]	30.00 [0.00, 142.50]	102.50 [0.00, 442.50]	.100
Post_CPB_CVP_AUT_16, min·mm Hg	0.00 [0.00, 26.88]	0.00 [0.00, 12.50]	30.00 [0.00, 120.00]	.015*
Post_CPB_CVP_AUT_20, min·mm Hg	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 20.00]	.004*
Postoperative data				
Hemoglobin, g/L	98.77 (16.17)	100.70 (16.49)	91.92 (13.08)	.016*
Neutrophil, %	90.15 [87.00, 92.30]	90.20 [87.00, 92.30]	89.90 [84.70, 92.30]	.500
Lymphocyte, %	4.30 [2.73, 6.83]	4.50 [2.80, 6.40]	3.90 [2.60, 8.60]	.981
NLR, %	21.37 [12.71, 33.48]	19.82 [13.59, 32.96]	23.00 [9.85, 35.62]	.978
VIS	12.00 [8.00, 22.00]	10.00 [7.00, 19.00]	27.00 [10.00, 32.00]	.001*

Data are presented as *n* (%), mean (standard deviation), or median (interquartile range). Variables with normal distribution are presented as mean (standard deviation); Variables without normal distribution are presented as median (interquartile range). MAP <55/60/65 time indicates the cumulative time of mean arterial pressure lower than 55/60/65 mm Hg; MAP_AUT_55/60/65 indicates the total area under curve below threshold mean artery blood pressure 55/60/65 mm Hg; CVP >12/16/20 time indicates the cumulative time of central venous pressure upper than 12/16/20 mm Hg; and CVP_AUT_12/16/20 indicates total area under curve above threshold central venous pressure 12/16 mm Hg.

ACC, aortic crossclamping; *AKI*, acute kidney injury; *ASA*, American Society of Anesthesiologists; *AST*, aspartate transaminase; *BMI*, body mass index; *CPB*, cardiopulmonary bypass; *CVP*, central venous pressure; *LAD*, left atrial anteroposterior dimension; *LVEF*, left ventricular ejection fraction; *NLR*, neutrophil to lymphocyte ratio; *NT-proBNP*, N-terminal pro-brain natriuretic peptide; *NYHA*, New York Heart Association; *PADP*, pulmonary artery diastolic pressure; *PASP*, pulmonary artery systolic pressure; *Scr*, serum creatinine; *TIA*, transient ischemic attack; *VIS*, vasoactive-inotropic score.

* Statistical significance ($P < .05$).

"xgboost 2.0.3," and " lightgbm 4.3.0" packages in Python 3.12.3 were used to construct all ML models.

We evaluated the performance of the predictive models mainly in terms of discrimination and calibration. The area under the receiver operating characteristic curve (AUROC), the area under the precision-recall (PR) curve (AUPRC), the precision, and the F1-score were used in combination to evaluate the discriminative ability. The Brier score and calibration curve were used to assess the calibration ability. The Brier score ranges from 0 to 1, with lower scores indicating better calibration, and the degree of overlap between the calibration curve and the diagonal also can reflect the model's calibration.[24] The optimal threshold of each model was determined by the Youden index (Youden index = sensitivity + specificity − 1), and all these evaluation metrics were calculated as an average using a 5-fold CV. The 5-fold CV randomly divides the dataset into 5 equal parts, alternating between 4 parts as training data and 1 part as validation data to estimate the model parameters and evaluate the performance.[25] The final performance of each model was assessed by computing the average of each evaluation metric using a 5-fold CV. This method facilitates an accurate evaluation of the performance of the constructed models and the selection of the optimal model.

Model interpretation

In this study, we employed the SHapley Additive exPlanations (SHAP) analysis to interpret the optimal prediction model we constructed.[26] The SHAP summary bar plot and SHAP summary scatter plot were employed to analyze the model's global output,

Table II
Collinearity analysis of the variables screened by univariate and LASSO analyses

Variables	Coefficient	VIF
Preoperative hemoglobin, g/L	−0.02170	1.359
Preoperative NT-proBNP, pg/mL		
Preoperative lymphocyte, %		
PASP, mm Hg	0.00501	1.033
PADP, mm Hg		
LAD, mm	0.01739	1.592
Right atrial long axis dimension, mm		
Surgery duration, min		
CPB duration, min		
ACC duration, min	0.01176	1.650
Intraoperative blood loss, mL		
Intraoperative urine output, mL/kg/h	−0.03468	1.164
Intraoperative red blood cell transfusion, units		
Intraoperative VIS		
MAP <65 time, min	0.00243	1.737
MAP <60 time, min		
MAP <55 time, min		
MAP_AUT_65, min·mm Hg		
MAP_AUT_60, min·mm Hg		
MAP_AUT_55, min·mm Hg		
CPB_MAP <65 time, min		
Post_CPB_MAP <65 time, min		
Post_CPB_MAP <60 time, min		
Post_CPB_MAP <55 time, min		
Post_CPB_MAP_AUT_65, min·mm Hg	0.00005	1.663
Post_CPB_MAP_AUT_60, min·mm Hg		
Post_CPB_MAP_AUT_55, min·mm Hg		
CVP_AUT_16, min·mm Hg		
Pre_CPB_CVP >12 time, min		
Pre_CPB_CVP_AUT_12, min·mm Hg		
Post_CPB_CVP >16 time, min		
Post_CPB_CVP >20 time, min	0.00103	1.284
Post_CPB_CVP_AUT_16, min·mm Hg		
Post_CPB_CVP_AUT_20, min·mm Hg		
Postoperative hemoglobin, g/L		
Postoperative VIS		

MAP <55/60/65 time indicates the cumulative time of mean arterial pressure lower than 55/60/65 mmHg; MAP_AUT_55/60/65 indicates the total area under curve below threshold mean artery blood pressure 55/60/65 mmHg; CVP >12/16/20 time indicates the cumulative time of central venous pressure upper than 12/16/20 mmHg; and CVP_AUT_12/16/20 indicates total area under curve above threshold central venous pressure 12/16 mmHg.
ACC, aortic crossclamping; *CPB*, cardiopulmonary bypass; *LAD*, left atrial anteroposterior dimension; *NT-proBNP*, N-terminal pro-brain natriuretic peptide; *PADP*, pulmonary artery diastolic pressure; *PASP*, pulmonary artery systolic pressure; *VIF*, variance inflation factor; *VIS*, vasoactive-inotropic score.

which can clarify the importance ranking of the features and their positive correlation with the results. Furthermore, the SHAP dependence plot was used to visualize the contribution of a specific feature to the prediction results at different values, which focuses more on the impact of individual features. The "shap 0.45.0″ package in Python 3.12.3 generated all SHAP plots. The best-performing model was further integrated into a risk web calculator for use.

Statistical analysis

All statistical analyses in this study were performed in R, version 4.2.2, and Python 3.12.3. First, the Shapiro–Wilk test was used to assess the normality of continuous variables. According to the Shapiro-Wilk test, continuous variables were described as the mean ± standard deviation or median ± interquartile range . Categorical variables were reported as counts (frequencies). Next, differences in continuous variables between the 2 groups were assessed using *t* tests for normally distributed data or Mann-Whitney *U* tests for non-normally distributed data. We used the χ^2 test or Fisher exact test to compare differences between the 2 groups, as appropriate for categorical variables.

Table III
The average performance of the 5 different ML models by 5-fold cross-validation

	AUROC	AUPRC	Precision	F1-score	Brier score
LR	0.897	0.787	0.520	0.616	0.120
XGB	0.869	0.727	0.493	0.636	0.126
RFC	0.893	0.813	0.746	0.698	0.104
LGBM	0.898	0.802	0.793	0.722	0.106
GNB	0.888	0.779	0.467	0.602	0.115

AUPRC, the area under the precision-recall (PR) curve; *AUROC*, the area under the receiver operating characteristic curve; *GNB*, Gaussian naive Bayes; *LGBM*, light gradient boosting machine; *LR*, logistic regression; *ML*, machine learning; *RFC*, random forest classifier; *XGB*, extreme gradient boosting.

Results

Baseline characteristics

Among 135 initially screened patients, 114 met the criteria and were enrolled in the final analysis. The study cohort had a median age of 52 years (interquartile range, 45–60), of whom 81.6% were male. Postoperative AKI (KDIGO stage 2–3) developed in 25 patients (21.9%). Patient characteristics are detailed in Table I.

Intraoperative hemodynamic variables and AKI

The univariate analysis revealed that both intraoperative hypotension and venous congestion were associated with an increased risk of postoperative AKI. The duration and AUT of hypotension at MAP thresholds of 65, 60, and 55 mm Hg assessed throughout the entire surgical procedure were found to be associated with the occurrence of postoperative stage 2–3 AKI. In contrast, the duration and AUT of venous congestion throughout the surgical procedure at CVP thresholds of 12, 16, and 20 mm Hg exhibited a weak correlation with the outcome incidence. According to a statistical significance threshold of $P < .05$ bilaterally, only the AUT above threshold CVP 16 mm Hg (CVP_AUT_16) demonstrated a statistically significant difference between the 2 groups.

Our stratified analysis of intraoperative hemodynamics across pre-CPB, CPB, and post-CPB phases demonstrated that post-CPB hypotension exhibited the strongest association with postoperative stage 2–3 AKI development, while venous congestion showed significant correlations both before and after CPB ($P < .05$, Table I). Multivariate analysis identified 3 independent hemodynamic risk factors (non-zero coefficient, Table II): the cumulative time of MAP lower than 65 mmHg (MAP <65 time), the AUT above the threshold MAP of 65 mm Hg after CPB (Post_CPB_MAP_AUT_65), and the cumulative time of CVP greater than 20 mm Hg after CPB (Post_CPB_CVP >20). These findings confirm that both intraoperative hypotension and venous congestion events increase the risk of postoperative AKI.

Model variable selection

A total of 36 variables were identified as being associated with the outcome by univariate analysis, with a P value < .05 (Table I). Subsequently, the LASSO algorithm identified 8 non-zero coefficient features related to the outcome, including Preoperative hemoglobin, PASP, Left atrial anteroposterior dimension, ACC duration,

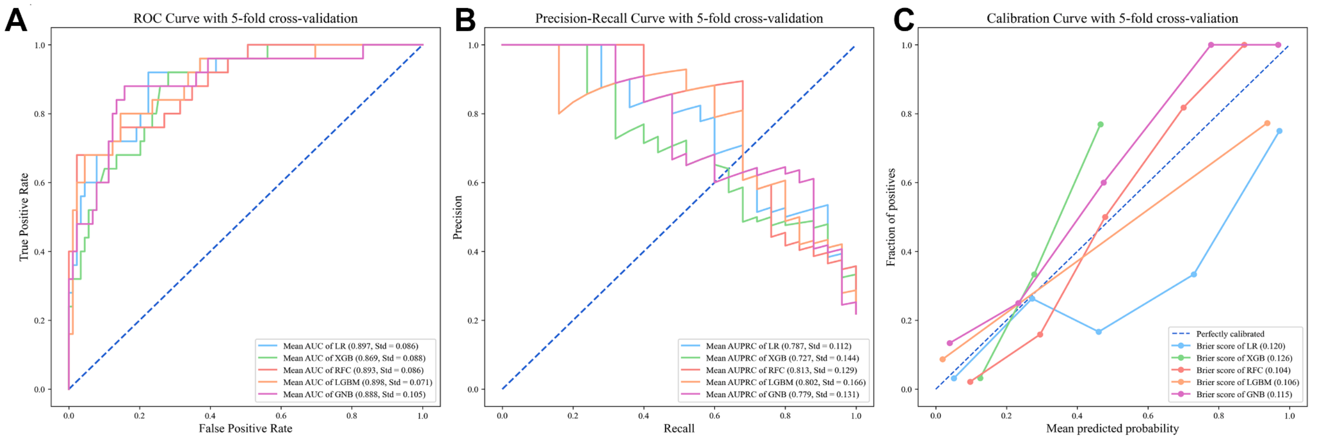

Figure 2. Performance of 5 machine-learning models. (A) AUROC, (B) AUPRC, and (C) Calibration curves of the machine-learning model for predicting stage 2−3 AKI with 5-fold cross-validation. *AUPRC*, the area under the precision-recall (PR) curve; *AUROC*, the area under the receiver operating characteristic curve; *GNB*, Gaussian naive Bayes; *LGBM*, light gradient boosting machine; *LR*, logistic regression; *RFC*, random forest classifier; *XGB*, extreme gradient boosting.

Intraoperative urine output, MAP <65 times, Post_CPB_MAP_AUT_65 and Post_CPB_CVP >20 times (Table II). Furthermore, the VIF test revealed no significant covariance between these 8 variables.

Model evaluation and comparison

The 8 aforementioned variables were used in constructing 5 ML models, and their predictive performance for the outcome was assessed using multiple metrics. The optimal hyperparameter combinations for each model were presented in Supplementary Table S1 and the detailed model performance under 5-fold CV is presented in Supplementary Table 2. The LGBM model with a 5-fold CV exhibited the greatest AUROC (0.898), AUPRC (0.802), precision (0.793), and F1-score (0.722) among the 5 ML models in predicting stage 2−3 AKI. Furthermore, the LGBM model exhibited optimal calibration ability, as indicated by a Brier score of 0.106. The results of the other models are presented in Table III and illustrated in Figure 2.

Model interpretation

As illustrated in Figure 3, we used SHAP analysis to evaluate the significance of each feature in the LGBM model and its impact on model prediction. Figure 3, *A*, illustrated that LAD, ACC duration, and preoperative hemoglobin were the 3 most significant features. Figure 3, *B*, demonstrated that all factors exhibited a positive correlation, except preoperative hemoglobin and intraoperative urine output, which showed a negative correlation with the risk of occurrence of the postoperative stage 2−3 AKI. This study also considered the subgroup-specific outcomes of variables in the population with or without postoperative stage 2−3 AKI (Figure 3, *C*).

Discussion

In this study, we thoroughly investigated the impact of perioperative hemodynamic factors on the development of postoperative stage 2−3 AKI in patients undergoing OHT while developing 5 ML prediction models incorporating other relevant perioperative risk factors. The LGBM model demonstrated optimal overall predictive performance (AUROC, 0.898; AUPRC, 0.802) and was consequently selected as the final risk prediction model for OHT-AKI. This is the first study to collectively incorporate 3 hemodynamic parameters—PASP, MAP, and CVP—into ML models for predicting AKI risk after OHT in adults.

The first advantage of this study is the development of a risk prediction model that showed potential in predicting the risk of developing Stage 2−3 AKI in patients after OHT. In addition, the construction of OHT-AKI models in the past mostly relied on traditional logistic regression methods.[11,13,27] Compared with conventional logistic regression, ML can identify better numbers of linear and nonlinear variables and time-varying variables related to clinical outcomes.[28] It is also worth noting that LGBM, an integrated learning algorithm based on a decision tree, is faster and more efficient than traditional algorithms and has been widely used in various medical prediction tasks.[29−31] Li et al[32] previously used the ML model to predict the risk of severe (stage 3) AKI after heart transplantation, with an AUC of 0.821. Our LGBM model was significantly superior to this model, possibly because it neglected to study the risk of developing moderate AKI. However, the hazard and risk of moderate AKI should not be ignored.[33] In clinical practice, moderate-to-severe (stage 2−3) AKI receives greater attention due to the complexity of its management.[34] Studies indicate that in complex cardiac surgery, patients without AKI and those with stage 1 AKI exhibit comparable long-term survival rates. By contrast, patients with AKI stage 2 or greater AKI show significantly greater in-hospital mortality than those without postoperative AKI.[35] The prediction model constructed in this study makes up for the gap in this field to a certain extent and provides a powerful tool for the comprehensive assessment of OHT-AKI risk.

Second, this study explored the hemodynamic factors at different periods during OHT surgery, especially the effects of hypotension and venous congestion on OHT-AKI. Most previous studies focused on the relationship between preoperative[11] and postoperative[36] partial hemodynamics such as MAP and the development and the development of OHT-AKI but failed to consider the impact of these parameters during surgery. However, intraoperative hemodynamic changes are an important index to evaluate renal hypoperfusion and guide treatment.[37] We divided the intraoperative period into 3 stages with the start and end of CPB as nodes, to explore the influence of hemodynamic changes on postoperative AKI at different periods during surgery (before CPB, during CPB and after CPB).[38] Such segmented analysis facilitates dynamic assessment of cardiac function and systemic conditions

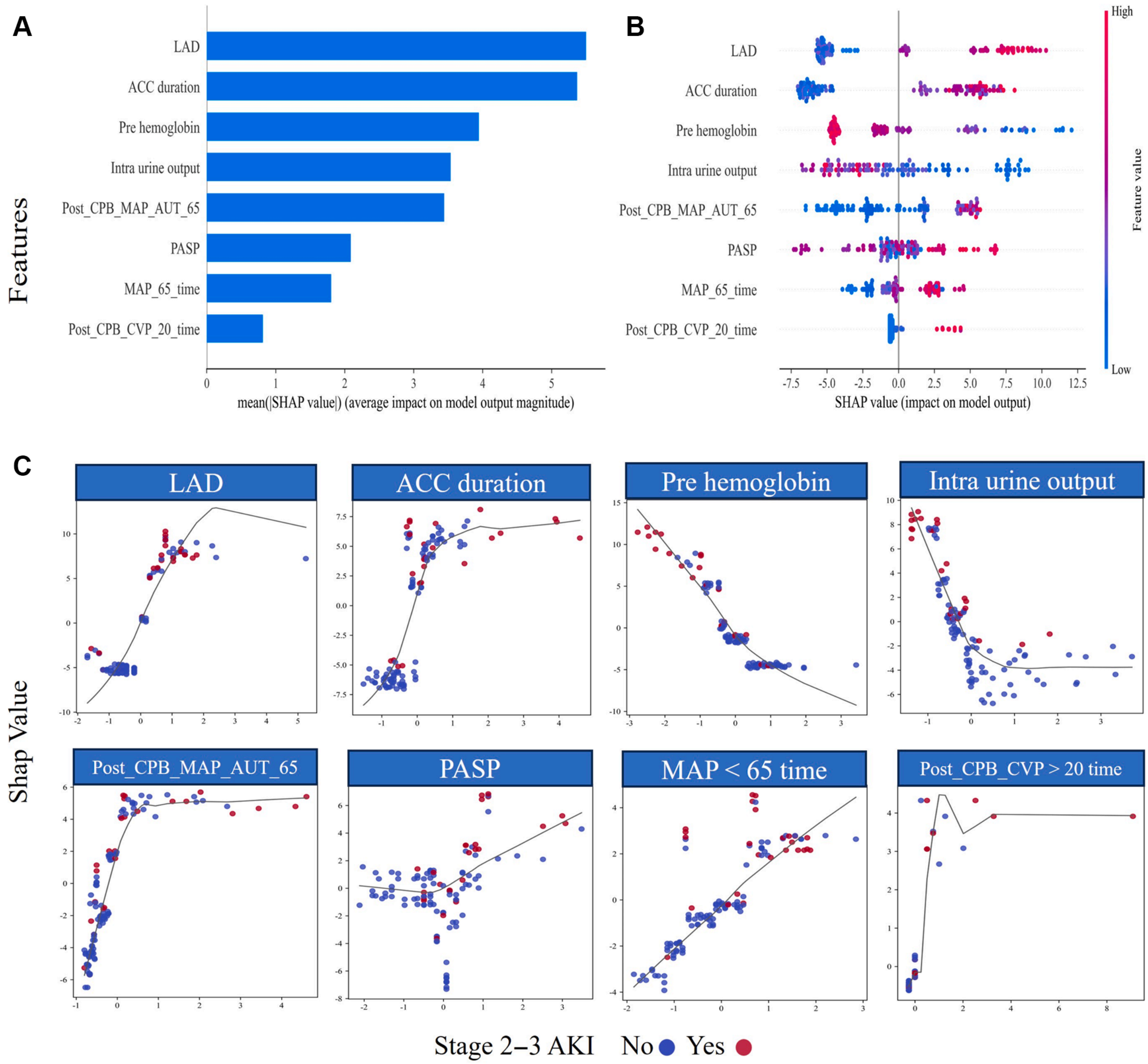

Figure 3. SHAP diagram of LGBM model. (A) SHAP value ranking of the variables in the model. (B) SHAP honeycomb diagram of the LGBM model. (C) SHAP dependence plot of variables in LGBM model. The x-axis represents the raw values of each variable, and the y-axis indicates the SHAP values of variables. MAP <65 time indicates the cumulative time of MAP lower than 65 mm Hg; Post_CPB_MAP_AUT_65, the AUT above the threshold MAP of 65 mm Hg after CPB; and Post_CPB_CVP >20, the cumulative time of CVP greater than 20 mm Hg after CPB. *ACC*, aortic crossclamping; *CPB*, cardiopulmonary bypass; *CVP*, central venous pressure; *LAD*, left atrial anteroposterior dimension; *LGBM*, light gradient boosting machine; *PASP*, pulmonary artery systolic pressure; *SHAP*, SHapley Additive exPlanations.

and provides important insights into hemodynamic management throughout surgery. It is also worth noting that, unlike previous studies that only set a single threshold or utilize single-point measurements of MAP or CVP to represent patients' intraoperative hemodynamics.[39–41] In this study, MAP and CVP time series data measured every 5 minutes were derived, and multiple thresholds were set to quantify venous congestion and hypotension as the duration and AUT of CVP above the threshold and MAP below the threshold. Thus, anesthesiologists can utilize this prediction to more effectively prevent postoperative AKI by reducing venous congestion and intraoperative hypotension.[41,42]

Third, for the first time in this study, PASP, MAP, and CVP were incorporated into the prediction model of OHT-AKI, which can more comprehensively reflect the perioperative hemodynamic state and its influence on the kidney. A single hemodynamic variable may not comprehensively reflect the circulatory status, and the development of AKI involves multiple complex and synergistic injury mechanisms.[43,44] Our study found that high preoperative PASP increased the risk of OHT-AKI, and previous studies have confirmed preoperative pulmonary hypertension (PH) as an independent risk factor for OHT-AKI, which is consistent with our findings.[45] Our multivariate analysis also showed that MAP <65 mm Hg duration during surgery, the AUT of MAP <65 mm Hg after CPB, and CVP >20 mmHg duration after CPB were independent risk factors for AKI after OHT. The decrease in MAP may be due to the initial dysfunction of the new heart after transplantation[46] and the complete loss of efferent and afferent autonomic nerve connections resulting in donor heart denervation resulting in abnormal blood flow regulation.[47,48] MAP is the driving force of renal perfusion, and decreased MAP can reduce renal perfusion and decrease GFR.[49,50] For CVP throughout surgery, only the duration of CVP >20 after CPB was an independent risk factor for OHT-AKI. Intraoperative CVP may be affected by venous blood volume, vascular tone, cardiac output, right ventricular compliance, and intrathoracic pressure, and intraoperative venous congestion is thought to occur only before or after CPB.[51] Moreover, patients with preoperative PH are considered to be at high risk of developing right ventricular failure (RVF) when CPB is discontinued.[52] RVF and PH can elevate the CVP.[45] The increase of CVP leads to elevated renal venous pressure, reducing the GFR.[11,51] More importantly, the kidneys are sensitive to hemodynamic changes, and the difference between arterial perfusion pressure and venous outflow pressure must be large enough to maintain adequate renal blood flow and glomerular filtration.[32] When low MAP and high CVP work together, the effective perfusion pressure of the kidney (MAP-CVP) is significantly reduced, further aggravating kidney injury. Therefore, maintaining normal hemodynamic parameters, preventing preoperative PH, closely monitoring intraoperative CVP and MAP changes, and timely intervention of abnormal conditions are essential to reduce the risk of AKI.

Finally, this study also found that preoperative hemoglobin, LAD, intraoperative urine output, and ACC duration were closely related to postoperative AKI development. Among them, a preoperative hemoglobin level that is too low will lead to reduced oxygen delivery to the kidney and cause tissue hypoxia. At the same time, the intraoperative urine output often means insufficient renal blood flow and renal perfusion pressure, which will promote the development of AKI.[3,53−55] Several studies have confirmed both as independent risk factors for AKI after cardiac surgery.[37,56] This study also showed that prolonged ACC during surgery significantly increases the risk of postoperative AKI. In the case of aortic crossclamping, immune activation, oxidative stress, and inflammatory responses may be promoted, leading to renal ischemia-reperfusion injury and postoperative AKI.[57] In addition, the study of Li et al[32] showed that the LAD is an important predictor of OHT-AKI, which is consistent with our findings. In short, actively identifying potential risk factors and improving perioperative management measures in clinical practice may help mitigate the progression of AKI.

Our study has some limitations that need to be addressed in future work. First of all, as a retrospective, single-center, observational study, our data collection was constrained by the completeness of electronic medical records. Several potential predictors—including Interagency Registry of. Mechanically Assisted Circulatory Support profiles, mechanical circulatory support details, immunosuppression regimens, and myocardial contractility indices—could not be incorporated into the analysis because of inherent limitations in retrospective data collection—a common challenge in such studies. Future prospective studies should implement structured data-collection protocols to capture these critical variables. Second, this study focused on preoperative and intraoperative factors. The absence of postoperative hemodynamic feature data may affect the model performance, although the current model has satisfactory predictive performance. In addition, there is a lack of independent data sets from other centers for external validation, and the model needs to be applied to a multicenter patient population in the future to verify its clinical utility. Finally, individual variation in baseline MAP and CVP values is common, and setting absolute thresholds may not apply to all patients.[16,58] Although we set multiple thresholds to minimize this error, the optimal threshold for each patient must be explored to optimize intraoperative hemodynamic management.

In conclusion, this study successfully developed and validated an ML-based predictive model integrating perioperative hemodynamic indices—specifically PASP, MAP, and CVP—to identify patients at risk of stage 2−3 AKI after OHT. By monitoring patients' fundamental hemodynamic parameters, clinicians can optimize perioperative medication regimens with greater precision, mitigate the occurrence and progression of postoperative AKI, ultimately improving patient prognosis.

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Conflict of Interest/Disclosure

The authors declare that they have no competing interests.

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Data availability

The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Xinlong Zhang: Writing − review & editing, Writing − original draft, Investigation, Data curation, Conceptualization. **Wanxia Li:** Writing − review & editing, Writing − original draft, Software, Methodology, Formal analysis. **Meijun Zhou:** Writing − review & editing, Writing − original draft. **Zhou Zhou:** Writing − review & editing, Supervision, Conceptualization. **Zhixiang Li:** Writing − review & editing, Data curation. **Rui Ding:** Writing − review & editing, Data curation. **Chen Chen:** Writing − review & editing. **Jinyi Shi:** Writing − review & editing. **Jialin Yin:** Writing − review & editing. **Hongwei Shi:** Writing − review & editing. **Jianjun Zou:** Writing − review & editing, Visualization, Supervision, Conceptualization. **Yanna Si:** Writing − review & editing, Supervision, Conceptualization.

Supplementary Material

Supplementary material associated with this article can be found, in the online version, at [https://doi.org/10.1016/j.surg.2025.109646].

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